Quality Vision Agent

Easy Explanation Notes for 3-Sector Gap Analysis

Purpose: This document explains the final gap analysis in simple meeting language, so you can confidently explain the logic to your manager without sounding like you are reading a technical report.

Question	Simple answer
What are we comparing?	The approved image prototype is the base. We compare it against our internal HTML prototype, public manufacturing usage examples, and competitor/third-party offerings.
What is the goal?	Not to criticize the approved prototype. The goal is to identify practical add-ons that make it more client-ready and market-ready.
What should we avoid?	Avoid saying companies expect something unless the source says they use it or offer it. Use source-backed language only.

1. One-Minute Summary You Can Say in the Meeting

Meeting opening line

"The approved image prototype is already a strong Quality Vision dashboard. Our analysis is not asking to redesign it. We compared it from three angles: our internal HTML prototype, manufacturing companies that have publicly used similar AI quality/vision solutions, and competitors/third-party vendors offering similar capabilities. Based on that, we identified a few practical add-ons like Executive Cockpit, filters, NCR/SAP/MES actions, live inspection strip, defect library, AI assistant, camera/edge health, and class-wise model performance."

2. How to Understand the Three Gap Sectors

Sector	What it means in simple words	What we are trying to prove
Sector 1: Approved prototype vs internal HTML prototype	We check what our own HTML prototype already had, but the approved image prototype does not clearly show.	These are low-risk add-ons because we already explored them internally.
Sector 2: Approved prototype vs manufacturing usage examples	We check what manufacturers are publicly saying they already use in AI quality inspection or visual inspection.	These show that the market is already using similar capabilities.
Sector 3: Approved prototype vs competitors / third-party providers	We check whether other companies are already offering similar visual inspection or quality intelligence solutions.	These show what our product may need to look competitive.

3. Sector 1 - Internal HTML Prototype vs Approved Image Prototype

This is the first gap your manager asked for. Here the source is not a public website. The source is our own internal HTML prototype compared with the leadership-approved image prototype.

HTML feature	What it means	Why we suggested it as a gap	Simple manager explanation
Executive Cockpit	A business-facing view showing impact like cost saved, labour saved, downtime avoided, and quality loss avoided.	Approved prototype is strong operationally, but a separate business impact view is not clearly visible.	"This helps leadership see business value, not only defect numbers."
Detailed filters	Filters such as plant, line, shift, time range, defect type, severity, and model version.	Approved prototype has plant context, but deeper investigation filters are not strong across screens.	"Filters make the dashboard useful for root-cause analysis."
Export NCRs / Sync MES /	Buttons/actions to export defect records and	Approved prototype mentions SAP QM, but	"This shows what happens after defect

SAP QM actions	sync with manufacturing systems.	direct workflow actions are not clearly visible.	detection."
Assign / Escalate / Create NCR workflow	Defects can be assigned to an owner, escalated, converted into NCR, and closed.	Approved defect log shows action/status, but not full ownership workflow.	"This converts AI detection into accountable business action."
Live Inspection Strip	A top strip showing cameras live, FPS, latency, parts inspected, and defects found.	Approved prototype has KPIs, but the real-time camera/inspection feel can be stronger.	"This proves the product is inspecting live production, not only showing reports."
Defect Library / NCR History	A history/archive of past defects, NCRs, closure status, and repeat issues.	Approved prototype has real-time defect log, but not a clear historical archive.	"This is needed for audit, traceability, and customer complaint tracking."
Quality Vision AI Assistant	A quality-specific AI chat inside Unified AI.	Approved prototype has Unified AI, but not a dedicated Quality Vision assistant experience.	"This helps managers ask questions instead of manually reading every chart."
Class-wise Model Performance	Performance by defect type, such as surface crack, dimensional tolerance, porosity, etc.	Approved ML page shows overall mAP/precision, but not performance for each defect type.	"Overall accuracy is not enough; clients will ask which defect classes are reliable."

4. Sector 2 - Manufacturing Companies: What They Are Using and How We Derived Gaps

This section is based only on what public sources say companies are using. We are not saying these companies are Infocepts customers.

Source	What the source actually says	How we converted it into our prototype gap	How to explain it simply
BMW Group - GenAI4Q	BMW says GenAI4Q gives tailored inspection recommendations and creates an individual inspection catalogue for vehicles.	Our prototype has insights, but a dedicated quality inspection recommendation/assistant flow is not clearly shown.	"BMW is using AI to guide inspection decisions. So we should strengthen our Quality Vision AI Assistant and recommendation panel."
Bosch - Industrial automation / AI	Bosch discusses AI in industrial automation, including real-time fault analysis, troubleshooting, and quality assurance support.	Our prototype shows line quality, but the root-cause/troubleshooting layer can be stronger.	"Bosch shows AI is used not just to display issues but to help understand faults. So we should add root-cause insights and filters."
Foxconn / Powerleader / Midea via Huawei	Huawei says Foxconn uses AI inspection for silicone grease and nameplate checks in smart PV controllers with above 99% accuracy. The same Huawei page mentions Midea using AI inspection for refrigerator-factory checks such as adjustable leg, ecolabel, logo, and condenser checks.	These are different inspection scenarios/classes. Our approved prototype shows overall model metrics but not performance by each defect/scenario.	"The source lists specific inspection cases. That is why we suggest class-wise or scenario-wise model performance."
Airbus / Accenture	Accenture says AI and computer vision were used to analyze video feeds in aircraft final assembly, detect manufacturing issues, timestamp steps, and use images/video.	Our prototype has a defect log, but stronger image/video evidence and timestamped inspection history would improve traceability.	"Airbus example shows visual evidence and event history matter. So we should add defect evidence viewer and timestamped event history."
Might Electronics / Advantech	Advantech says Might Electronics uses PowerArena AI visual inspection with cameras at workstations, real-time abnormality alerts, visual records, and production history for tracing customer complaints.	Our prototype has current defect log, but a full defect history/archive is not clearly visible.	"This directly supports our Defect Library/NCR History gap because they use production history for traceability."

5. Example: How to Explain the Foxconn / Midea Row Clearly

Why did we say "class-wise model performance"?

The Huawei source does not use our exact phrase "class-wise model performance." That is our gap-analysis interpretation.
The source lists specific inspection scenarios: silicone grease, nameplate, adjustable leg, ecolabel, logo, condenser checks.
Each of these is a different type of inspection/defect scenario. In our approved prototype, we show overall mAP and precision, but we do not clearly show reliability for each defect/scenario.
So the safe explanation is: "Because manufacturers are using AI for multiple specific inspection scenarios, our prototype should show model performance and evidence for each defect/scenario, not only one overall score."

6. Sector 3 - Competitors and Third-Party Offerings: What They Offer and What Gap We Identified

This section checks whether other companies are already offering similar Quality Vision / AI visual inspection capabilities. Here we compare their public offering with our approved prototype.

Company	What they publicly offer	Gap in our approved prototype	Simple manager explanation
Wipro InspectAI	AI/computer-vision visual inspection with workflow, reporting/corrections, compliance, and ROI positioning.	Our prototype needs stronger Executive Cockpit, workflow actions, reporting/export, and compliance positioning.	“Wipro is positioning visual inspection with business impact and workflow, so we should show ROI and NCR/SAP actions clearly.”
LandingAI / LandingLens	Computer vision for manufacturing quality, Good/No Good classification, and specific defect examples like missing part, scratch, smudge.	Our prototype needs clearer defect taxonomy, defect evidence, and class-wise model performance.	“LandingAI shows defect-specific inspection. So our model page should show defect-class performance, not just overall mAP.”
IBM Maximo Visual Inspection	AI-assisted workflows across mobile devices, cameras, drones, and edge environments; also mentions real-time defect detection and root-cause analysis.	Our prototype should strengthen camera/edge health, workflow closure, and root-cause recommendations.	“IBM emphasizes deployment and workflow. So we should show camera/edge status and defect closure flow.”
KEYENCE / Cognex	Machine vision systems for automated defect detection, flaw detection, cameras, and real-time inspection credibility.	Our prototype references cameras, but live camera/FPS/device health is not prominent.	“Vision vendors make the live inspection setup very visible. We should add live inspection strip and camera health.”
SAP Digital Manufacturing Visual Inspection	SAP documentation describes visual inspection for identifying nonconformances early in manufacturing.	Approved prototype mentions SAP QM but does not show nonconformance/SAP/MES lifecycle actions.	“Since SAP connects inspection to nonconformance, we should show Create NCR, Send to SAP QM, Sync MES, and status tracking.”
Siemens Inspekto	AI-based visual inspection offering focused on easy setup and inspection readiness.	Our prototype has model metrics but not deployment/onboarding readiness or integration status.	“Siemens highlights ease of setup. We can show model onboarding and deployment readiness.”
Mu Sigma	Adjacent analytics/AI competitor evidence: Azure AI/ML, MLOps, MuNLQ natural-language query, digital twin, reporting.	Not a direct Quality Vision Agent. But it supports AI assistant, reporting, MLOps/model governance gaps.	“Do not say Mu Sigma offers the same Quality Vision product. Say their analytics/AI positioning supports our AI assistant and MLOps add-ons.”

7. Source Verification: Where to Check in Each Link

Use Ctrl+F inside each source page. These search words are what connect the source to the gap. This will help you answer confidently if your manager checks link by link.

Source	Ctrl+F words	Link
BMW Group	tailored inspection recommendations; 1,400 vehicles; individual inspection catalogue	Open source
Bosch	fault analysis and troubleshooting; quality assurance; DeepInspect; AI Analytics Platform	Open source
Huawei / Foxconn / Midea	silicone grease; nameplates; above 99%; Midea; adjustable leg; ecolabel; condenser	Open source
Accenture / Airbus	video feeds; detect manufacturing issues; timestamped; images and video	Open source
Advantech / Might Electronics	cameras set up at each workstation; abnormality; production history; exact assembly operation	Open source
Wipro InspectAI	visual inspection; Streamline Inspection Workflow; reporting; compliance; Maximize ROI	Open source
LandingAI	Good / No Good; Part Missing; Scratch; Smudge; upload; label; Train	Open source
IBM Maximo Visual Inspection	AI-assisted workflows; cameras; edge environments; real-time defect detection; root cause analysis	Open source
KEYENCE	AI and rule-based vision tools; scratches, stains, or foreign material; flag defects in real time	Open source
Cognex	automated defect detection; machine vision; improve product quality; operational efficiency	Open source
SAP Visual Inspection	Visual Inspection; nonconformances; manufacturing; SAP Digital Manufacturing	Open source
Siemens Inspekto	AI-based visual inspection; no AI expertise; quality; inspection	Open source

Mu Sigma	Azure AI/ML; MLOps; MuNLQ; Digital Twin; interactive reporting	Open source

8. How the Main Add-ons Were Decided

Recommended add-on	Why we recommended it	Which sector supports it
Executive Cockpit	Needed to show ROI/business value, not only quality metrics.	Internal HTML + Wipro + Mu Sigma
Plant/Line/Shift/Time filters	Needed for root-cause investigation.	Internal HTML + Bosch
Export NCR / Sync MES / SAP QM	Needed to show what happens after defect detection.	Internal HTML + SAP + Wipro
Assign/Escalate/Create NCR workflow	Needed to show ownership and closure.	Internal HTML + Wipro + IBM
Live Inspection Strip	Needed to prove inspection is live with cameras/FPS/latency.	Internal HTML + KEYENCE + Cognex + IBM
Defect Library / NCR History	Needed for audit, historical tracking, and customer complaint traceability.	Internal HTML + Advantech + Airbus
Quality Vision AI Assistant	Needed for natural-language questions and inspection recommendations.	Internal HTML + BMW + Mu Sigma
Class-wise Model Performance	Needed to show reliability by defect/scenario.	Internal HTML + Huawei/Foxconn/Midea + LandingAI

9. What Not to Say in the Meeting

Avoid saying	Say this instead
"All these companies are our customers."	"These are public manufacturing references; we are using them only as evidence of what the industry is already using."
"Carrier is a focus customer."	"Carrier is not positioned as a focus customer in this document."
"Mu Sigma offers the same Quality Vision Agent."	"Mu Sigma is an adjacent analytics/AI competitor reference, useful for AI assistant, reporting, and MLOps positioning."
"The source says class-wise model performance."	"The source lists multiple defect/inspection scenarios, so we derived the need to show class-wise or scenario-wise performance."
"The approved prototype is weak."	"The approved prototype is strong as a base; the gaps are enhancement areas for client-readiness."

10. Final Short Script for You

Use this during your explanation

"We compared the approved Quality Vision prototype from three angles. First, against our internal HTML prototype, where we found productisation add-ons like Executive Cockpit, filters, Export NCR, Sync MES, Defect Library, live inspection strip, and AI assistant. Second, against public manufacturing usage examples like BMW, Bosch, Huawei/Foxconn/Midea, Airbus, and Advantech, where companies are already using AI for inspection recommendations, visual inspection, traceability, and production history. Third, against competitors and third-party offerings like Wipro InspectAI, LandingAI, IBM Maximo Visual Inspection, KEYENCE, Cognex, SAP, Siemens, and Mu Sigma. Based on these three angles, our recommendation is not to redesign the approved prototype, but to enhance it with practical features that improve business value, workflow integration, traceability, live inspection credibility, and AI trust."

11. If Manager Asks “Who Is Doing Which One Better?”

Area	Who is stronger / evidence source	What we should learn
Business value / ROI	Wipro InspectAI, internal HTML Executive Cockpit	Add Executive Cockpit and business-impact metrics.
Inspection recommendations	BMW GenAI4Q	Strengthen Quality Vision AI Assistant and recommendation logic.
Root-cause investigation	Bosch	Add stronger filters and root-cause explanations.
Scenario-specific defect inspection	Huawei/Foxconn/Midea and LandingAI	Add class-wise/scenario-wise performance.
Traceability/history	Advantech/Might Electronics and Airbus/Accenture	Add Defect Library, timestamped event history, and image/video evidence.
Live camera/edge credibility	KEYENCE, Cognex, IBM	Add Live Inspection Strip and Camera/Edge Health.
SAP/MES/nonconformance workflow	SAP and Wipro	Add NCR/SAP/MES lifecycle actions.